

Experiment No: 0 Pre-requisites

Title: 8051 – Study of different types of arithmetic and logical instructions

Class: T.E.

A.Y.:

Semester: I

Roll No.:

Name:

Date of performance:

Date of Submission:

Signature:

AIM: Study of different types of arithmetic and logical instructions.

1. WAP to add and subtract two 8-bit numbers.
2. WAP to multiply and divide two 8-bit numbers.
3. WAP to add and subtract two 16-bit numbers.
4. WAP to add two BCD numbers.
5. WAP to convert a 8-bit packed BCD to an unpacked BCD number.
6. WAP to count the number of zeros and ones in an 8-bit data.

S/W AND H/W TOOLS: 8051 IDE

ALGORITHMS**1) Addition and subtraction of two 8-bit numbers**

- i. Move first 8-bit number to Accumulator
- ii. Move second 8-bit number to B Register
- iii. Add numbers in A and B register and store the result in A.
- iv. Copy the result in R1 register.
- i. Move first 8-bit number to Accumulator
- ii. Move second 8-bit number to B Register
- iii. Subtract the numbers in A and B register and store the result in A.
- iv. Copy the result in R2 register.

2) Multiplication and division of two 8-bit numbers.

- i. Move first 8-bit number to Accumulator
- ii. Move second 8-bit number to B Register
- iii. Using MUL instruction multiply contents in A and B
- iv. Store the result in register R3.
- i. Move the Dividend to Accumulator
- ii. Move the Divisor to B Register
- iii. Using DIV instruction divide contents in A by B
- iv. Store the Quotient in register R1 and Remainder in R2

3) Addition and subtraction of two 16-bit numbers

- i. Move LSB byte of first 16-bit number to Accumulator
 - ii. Move LSB byte of second 16-bit number to R1 Register
 - iii. Perform addition using ADD instruction. Monitor the PSW (flag register)
 - iv. Store the added result in some temporary register.
 - v. Move MSB byte of first 16-bit number to Accumulator
 - vi. Move MSB byte of second 16-bit number to R2 Register
 - vii. Perform addition using ADDC instruction.
 - viii. Store the result in any temporary register.
-
- i. Clear the contents of Carry flag
 - ii. Move LSB byte of first 16-bit number to Accumulator
 - iii. Move LSB byte of second 16-bit number to R1 Register
 - iv. Perform subtraction using SUBB instruction. Monitor the PSW (flag register)
 - v. Store the result in some temporary register.
 - vi. Move MSB byte of first 16-bit number to Accumulator
 - vii. Move MSB byte of second 16-bit number to R2 Register
 - viii. Perform subtraction using SUBB instruction
 - ix. Store the result in any temporary register.

4) Addition of two BCD numbers

- i. Move first 8-bit number to Accumulator
- ii. Move second 8-bit number to B Register
- iii. Add numbers in A and B register.
- iv. Decimal adjust using DA A instruction
- v. Store the result in any temporary register

5) Convert a 8-bit packed BCD to an unpacked BCD number

- i. Move first 8-bit number to Accumulator
- ii. Mask the lower nibble of 8-bit data using data 0FH and ANL instruction
- iii. Store result in temporary register R1

- iv. Mask the upper nibble of 8-bit data using data 0F0H and ANL instruction
- v. Use SWAP instruction
- vi. Store result in temporary register R2

6) Count the number of zeros and ones in an 8-bit data.

- i. Move first 8-bit number to Accumulator
- ii. Declare a counter
- iii. Check ACC.0 bit
- iv. If the bit is 0 then increment contents in register R2, else increment contents in register R3.
- v. Rotate the contents in A using RR instruction.
- vi. Use DJNZ instruction to go to step (ii)

PROGRAMS

Attach 1-6 programs with output wherever possible.

CONCLUSIONS
